

Encrypting Partitions Using LUKS

Sensitive data needs total protection. And there's no better way of protecting your sensitive data than by encrypting it. This article is a tutorial on how to encrypt your laptop or server partitions using LUKS.



Sensitive data on mobile systems such as laptops can get compromised if they get lost, but this risk can be mitigated if the data is encrypted. Red Hat Linux supports partition encryption through the Linux Unified Key Setup (LUKS) on-disks-format technology. Encrypting partitions is easiest during installation but LUKS can also be configured post installation.

Encryption during installation

When carrying out an interactive installation, tick the *Encrypt* checkbox while creating the partition to encrypt it. When this option is selected, the system will prompt users for a passphrase to be used for decrypting the partition. The passphrase needs to be manually entered every time the system boots.

When performing automated installations, Kickstart can create encrypted partitions. Use the `--encrypted` and `--passphrase` option to encrypt each partition. For example,

the following line will encrypt the `/home` partition:

```
# part /home --fstype=ext4 --size=10000 --onpart=vda2
--encrypted --passphrase=PASSPHRASE
```

Note that the passphrase, `PASSPHRASE` is stored in the Kickstart profile in plain text, so this profile must be secured. Omitting the `--passphrase` option will cause the installer to pause and ask for the passphrase during installation.

Encryption post installation

Listed below are the steps needed to create an encrypted volume:

1. Create either a physical disk partition or a new logical volume.
2. Encrypt the block device and designate a passphrase, by using the following command: